Software Corp.

Project Management Deliverables: Smart Parking Platform

Course: CIS 4374 – Managing Teams and Stakeholders

Date: October 29, 2026

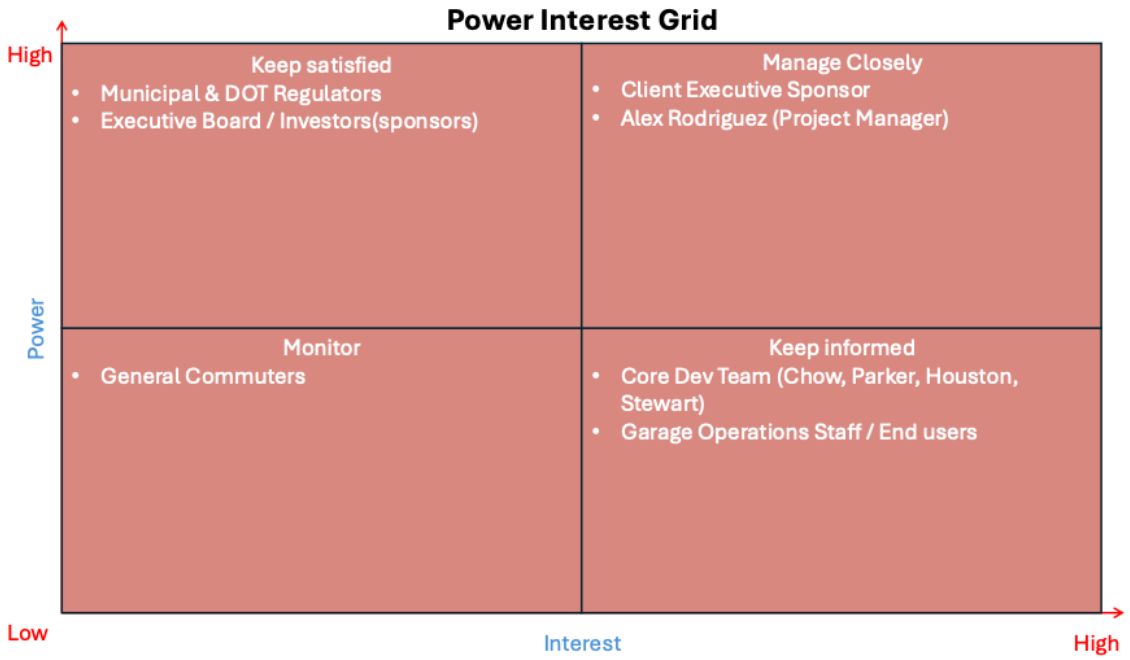
Document Scope: Stakeholder Analysis, Project Charter, and Team Charter

1. Stakeholder Analysis: Power–Interest Grid

1.1 Overview and Purpose

The Power–Interest Grid maps project stakeholders based on their level of organizational authority and their active stake in outcomes regarding the project. This categorization determines communication channels, reporting frequency, and escalation procedures across the 16-week project lifecycle.

1.2 Grid Layout



1.3 Quadrant Definitions and Engagement Protocols

Manage Closely (High Power, High Interest):

- Client Executive Sponsor (City Transit & Parking Authority): Holds final budget authorization and sign-off rights. Engaged through bi-weekly milestone demonstrations and executive summary variance reports.
- Alex Rodriguez (Project Manager): Responsible for schedule adherence, cross-functional alignment, and barrier removal. Manages the team scrums daily and leads status reviews.

Keep Satisfied (High Power, Low Interest):

- Municipal & Department of Transportation (DOT) Regulators: Ensure sensor safety certifications, ADA parking space compliance, and payment data security standards (PCI-DSS). Engaged via structured compliance checkpoints before release gates.
- Software Corp. Executive Board & Investors: Monitor portfolio return on investment, overhead costs, and delivery risks. Informed through monthly high-level dashboard summaries.

Keep Informed (Low Power, High Interest):

- Core Engineering Team (David Chow, Sarah Parker, Marcus Houston, Leonardo Stewart: Design, construct, and validate platform architecture. Engaged through daily standups, GitHub pull requests, and Trello backlog tracking.
- Garage Operations Staff & Attendants: End-operator users responsible for physical stall maintenance and system override actions. Engaged through workflow interviews, user acceptance testing (UAT), and operational runbooks.

Monitor / Minimum Effort (Low Power, Low Interest):

- General Commuters & Drivers: Day-to-day app users who rely on accurate telemetry to reserve and navigate to available stalls. Monitored post-launch through system error metrics and general app feedback.

2. Project Charter

2.1 Project Identification

Project Name: Smart Parking

Platform Organization: Software Corp.

Project Manager: Alex Rodriguez

Project Sponsor: City Transit & Parking Authority

Project Duration: 16 Weeks (1,080 Total Labor Hours Allocated)

2.2 Business Case & Purpose

Urban parking operators encounter operational blind spots and dips in revenue due to manual monitoring and inefficient driver stall searches. The Smart Parking Platform replaces manual oversight by streaming automated spot occupancy telemetry into a unified management hub, optimizing space utilization and driver throughput.

2.3 Project Objectives

- Deliver an operational web dashboard featuring sub-second WebSocket updates for stall availability.
- Construct an intuitive driver reservation interface built on Vue.js.
- Establish a resilient Node.js/Express backend integrated with relational database storage and cloud infrastructure.
- Complete full user acceptance testing (UAT) and a pilot deployment across 250 parking bays within 16 weeks.

2.4 Scope Boundaries

In-Scope:

- Development of real-time garage map layout and occupancy APIs.
- Spot reservation, automated billing calculations, and attendant manual override dashboards.
- Cloud infrastructure provisioning, monitoring, and automated database backups.

Out-of-Scope:

- Physical manufacturing or hardware repair of ultrasonic/optical bay sensors.
- Municipal traffic light network integration.

2.5 Key Milestone Schedule

Milestone 1 (Week 4): Project requirements approval, system architecture baseline, and database schema finalization.

Milestone 2 (Week 8): Core REST API endpoints and WebSocket telemetry stream functional.

Milestone 3 (Week 12): Front-end UI integration complete; initial security and load testing signed off.

Milestone 4 (Week 16): User acceptance testing (UAT) complete, pilot installation verified, and project closeout.

2.6 Project Governance & Approval Authority

Alex Rodriguez is authorized to allocate resources, manage sprint backlogs, and direct technical activities within the approved budget and schedule limits. Material changes to scope, schedule, or budget require formal written sign-off from the Client Executive Sponsor.

3. Team Charter

3.1 Mission and Purpose

The project team of Software Corp. is dedicated to engineering an enterprise-grade Smart Parking Platform on schedule and within budget, adhering to high standards of code quality, reliable testing, and transparent communication.

3.2 Team Composition and Roles

- **Alex Rodriguez – Project Manager:** Oversees project schedule, facilitates sprint scrums, coordinates stakeholder communications, and resolves blockers.
- **David Chow – Backend Engineer:** Designs and develops REST API services, WebSocket streaming handlers, and backend business logic.
- **Sarah Parker – Frontend Engineer:** Leads client-side architecture in Vue.js, component styling, and driver/operator interface workflows.
- **Marcus Houston – QA & Security Specialist:** Builds automated unit/integration test suites, runs penetration audits, and validates system stability.
- **Leonardo Stewart – Database & DevOps Architect:** Manages database schemas, data integrity, CI/CD pipeline automation, and cloud deployments.

3.3 Operating Principles and Working Agreements

Daily Scrum: 15-minute daily sync at 9:30 AM to discuss yesterday's progress, current commitments, and existing blockers.

Primary Communications: Asynchronous collaboration occurs in dedicated Slack/Teams channels. All formal tasks and sprint statuses must be maintained in Trello/Jira.

Version Control & Quality Standards:

- All work is developed on designated feature branches; direct commits to main are prohibited.
- Every pull request (PR) requires passing automated CI tests and at least one peer code review before merge.

3.4 Decision-Making Framework

Technical design decisions will be established through consensus between the domain lead and the system architect. If an impasse occurs, the Project Manager will evaluate delivery timelines and project objectives to determine the resolution.

3.5 Three Step Conflict Resolution Process

1. **Direct Peer Discussion:** Disagreeing parties must address conflicts directly and professionally within 24 hours.
2. **Facilitated Mediation:** If unresolved after 48 hours, the matter is brought to the Project Manager for formal mediation.
3. **Escalation / Architectural Review:** If technical disagreement impacts delivery scope or milestone integrity, a formal majority vote by technical leads is conducted to produce a binding decision.